

Module 9 – Big Ideas: Capstone Part 1, Data Collection, Wrangling, and Exploratory Data Analysis

Big Idea 1: The Capstone Is a Full Data Analyst Workflow

Module 9 begins the IBM Data Analyst capstone.

You are not just completing separate labs.

You are acting as an Associate Data Analyst working through a real analytics pipeline:

Collect data

Clean and prepare it

Explore patterns

Build evidence for dashboards

Prepare insights for stakeholders

The dataset is based on the Stack Overflow Developer Survey, which captures how developers use technologies today and what they want to learn in the future.

Key idea:

This module is about moving from course exercises to a complete analytics project.

Why it matters

Real data analysis is not one notebook or one chart.

It is a sequence of connected decisions that turn messy data into trustworthy insights.

Big Idea 2: Raw Data Is Not Dashboard-Ready

The main purpose of Part 1 is to prepare the data for later visualization and presentation.

Before you can build dashboards, you need to understand and clean the dataset.

Raw survey data can contain:

Duplicates
Missing values
Inconsistent categories
Skewed numeric fields
Outliers
Columns that need transformation

Key idea:

Dashboards are only as good as the data underneath them.

Why it matters

If the dataset is messy, the final charts may look polished but still tell the wrong story.

Module 9 gives you the foundation for stronger Module 10 dashboards and the final capstone presentation.

Big Idea 3: Data Collection Starts With Understanding the Web

The first part of Module 9 reviews how data is collected from external sources.

This includes:

HTTP requests
URLs
Request and response cycles
Status codes
Headers
Web content
APIs
HTML pages

Python's `requests` library is used to retrieve data from web resources.

Key idea:

Before analyzing data from the internet, you need to understand how that data is requested and returned.

Why it matters

Many real analytics projects begin outside your local machine.

Data may come from APIs, websites, files, databases, or third-party systems.

Knowing how to collect it is the first step in building an analytics pipeline.

Big Idea 4: APIs Provide Structured Data

APIs are one of the cleanest ways to collect data.

Instead of manually downloading files, an API lets you request structured data programmatically.

Common API formats include:

JSON

CSV

Excel exports

Database-backed endpoints

In Module 9, API work connects directly to collecting job and technology-related data.

Key idea:

APIs allow analysts to collect repeatable, structured data from external systems.

Why it matters

APIs are common in business analytics, product analytics, sports analytics, finance, marketing, and operations.

They make data collection more scalable and reproducible than manual downloads.

Big Idea 5: Web Scraping Is Useful When Data Lives in HTML

Not all data is available through clean APIs.

Sometimes the data you need is embedded in a web page.

That is where web scraping comes in.

In Module 9, web scraping includes:

Downloading HTML with `requests`

Parsing pages with BeautifulSoup

Finding tags and attributes

Extracting links

Extracting images

Extracting tables

Saving scraped data to CSV

Key idea:

Web scraping turns information from websites into structured data for analysis.

Why it matters

Analysts often need to collect data from imperfect sources.

Scraping is not always the first option, but it is an important tool when data is visible on the web but not packaged cleanly.

Big Idea 6: Exploration Comes Before Cleaning

Before changing the dataset, you need to understand it.

The first step is usually basic inspection:

`head()`

`info()`

`describe()`

`shape`

Column names

Data types

Missing value counts

Value counts

Key idea:

Do not clean data blindly.

First, inspect the dataset and understand what each column represents.

Why it matters

Cleaning decisions depend on context.

A missing value might mean “unknown,” “not applicable,” “not answered,” or “zero,” depending on the column.

A data analyst needs to understand the data before deciding what to change.

Big Idea 7: Duplicates Can Distort Results

Duplicate rows can make patterns look stronger than they really are.

In survey data, duplicates may affect:

- Counts

- Percentages

- Averages

- Technology popularity rankings

- Compensation summaries

- Dashboard metrics

Module 9 includes finding duplicates, analyzing duplicate patterns, and removing them carefully.

Key idea:

Duplicate records can quietly bias your analysis.

Why it matters

If duplicate responses remain in the dataset, your final dashboard may overcount certain groups or trends.

Removing duplicates is one of the first steps toward trustworthy analysis.

Big Idea 8: Missing Values Require Defensible Choices

Missing data is normal in real-world datasets.

In the Stack Overflow survey, some respondents may skip questions about:

- Education

- Remote work

Compensation
Job satisfaction
Experience
Technology preferences

Common strategies include:

Dropping missing values
Filling categorical values with the mode
Filling numeric values with the median or mean
Creating “Unknown” categories
Documenting the decision

Key idea:

Handling missing values is not just a technical step. It is an analytical judgment.

Why it matters

Different missing-value strategies can lead to different conclusions.

A good analyst chooses a method that fits the column, the business question, and the downstream analysis.

Big Idea 9: Normalization and Transformation Make Data Easier to Analyze

Some numeric fields, especially compensation, can be highly skewed.

A small number of very high values can distort averages and visualizations.

Module 9 introduces techniques such as:

Min-Max scaling
Z-score standardization
Log transforms
Forward fill
Feature scaling

Key idea:

Transformation helps make messy numeric data easier to compare, model, and visualize.

Why it matters

Without normalization or transformation, charts can be dominated by extreme values.

A good transformation can make patterns easier to see without deleting important information.

Big Idea 10: Feature Engineering Turns Raw Columns Into Useful Signals

Sometimes the original columns are not enough.

Analysts often create new variables to make the data more useful.

Examples include:

- Grouping years of experience into experience levels

- Creating cleaned country or region labels

- One-hot encoding categorical variables

- Creating simplified employment categories

- Creating transformed compensation fields

Key idea:

Feature engineering turns raw data into analysis-ready variables.

Why it matters

The best insights often come from thoughtful derived columns, not just the original dataset.

This is where domain understanding and technical skill come together.

Big Idea 11: EDA Helps You Find the Story Before the Dashboard

Exploratory Data Analysis, or EDA, is where you start looking for patterns.

In Module 9, EDA includes:

- Distributions

- Histograms

KDE plots
Box plots
Value counts
Grouped summaries
Cross-tabs
Heatmaps
Correlation analysis
Scatter plots

Key idea:

EDA helps you discover what is interesting before you decide what to present.

Why it matters

Dashboards should not be random collections of charts.

EDA helps you decide which metrics, comparisons, and trends are worth showing to stakeholders.

Big Idea 12: Outliers Can Reveal Problems or Important Stories

Outliers are values that are unusually high, low, or different from the rest of the data.

In Module 9, outlier analysis focuses especially on compensation.

Common methods include:

Box plots
Standard deviation thresholds
IQR bounds
Cleaned DataFrames with extreme values removed

Key idea:

Outliers should be investigated, not automatically deleted.

Why it matters

Some outliers are data errors.

Others are real and meaningful.

A strong analyst knows how to identify them, evaluate them, and decide whether they should remain in the analysis.

Big Idea 13: Correlation Helps Identify Relationships, Not Causation

Correlation analysis helps measure relationships between numeric variables.

Module 9 includes:

- Pearson correlation
- Spearman correlation
- Correlation heatmaps
- Scatter plots

Possible relationships include:

- Experience and compensation
- Work experience and job satisfaction
- Compensation and satisfaction
- Technology usage and respondent characteristics

Key idea:

Correlation can reveal relationships, but it does not prove cause and effect.

Why it matters

Stakeholders may be tempted to interpret every relationship as causal.

A good data analyst communicates correlation carefully and avoids overstating conclusions.

Big Idea 14: Technology Trends Are the Business Question

The capstone is not just about cleaning data.

The larger business question is:

What technologies are developers using now, and what do they want to use in the future?

This includes trends in:

- Programming languages
- Databases
- Platforms
- Frameworks
- Developer demographics
- Remote work
- Education
- Employment
- Compensation
- Satisfaction

Key idea:

The technical work supports a business story about technology trends.

Why it matters

Stakeholders do not only want clean data.

They want insight into where the technology landscape is heading.

Big Idea 15: Part 1 Feeds Part 2 and Part 3

Module 9 is Part 1 of the capstone.

It prepares the data for the next phases.

Part 1:

Collect, wrangle, and explore the data.

Part 2:

Create dashboards in IBM Cognos Analytics or Google Looker Studio.

Part 3:

Present findings in a PowerPoint report.

Key idea:

Each phase builds on the previous one.

Why it matters

If Part 1 is weak, the dashboards and final presentation will also be weak.

A clean, explored dataset makes the rest of the capstone much easier.

Big Idea 16: The Real Skill Is Building a Repeatable Analysis Process

The most important takeaway from Module 9 is not one Pandas command.

It is the full workflow:

- Collect data
- Inspect structure
- Remove duplicates
- Handle missing values
- Normalize or transform fields
- Engineer useful features
- Explore distributions
- Find outliers
- Analyze correlations
- Export clean data
- Prepare for dashboards

Key idea:

The goal is not to memorize every method.

The goal is to build a repeatable process for turning raw data into reliable insight.

Why it matters

This is what data analysts do in real work.

They do not just make charts.

They build trust in the data, find the story, and communicate it clearly.